

The JHU Machine Translation Systems for WMT 2018

Anonymous EMNLP submission

Abstract

We report on the efforts of the Johns Hopkins University to develop neural machine translation systems for the shared task for news translation organized around the Conference for Machine Translation (WMT) 2018. We developed systems for German–English, English–German, and Russian–English. Our novel contributions are iterative back-translation and fine-tuning on test sets from prior years.

1 Introduction

We carried out two relatively independent efforts on German–English language directions and Russian–English, using the Marian and Sockeye neural machine translation toolkits, respectively.

The German–English systems outperformed last year’s best result (37.0 vs. 35.1 (+1.9) for German–English, 29.1 vs. 28.3 (+0.8) for English–German), but fell short against this year’s best performing systems (45.3 vs. 48.4 (-3.1) and 43.4 vs. 48.3 (-4.9), respectively)¹. The best models this year used the Transformer model instead of the recurrent neural networks that our models are based on. Our novel contributions are iterative back-translation and fine-tuning on prior test sets.

For Russian–English, we carried out extensive hyperparameter search, with different numbers of layers, embedding and hidden state sizes, and drop-out settings.

2 German–English and English–German

The systems for the German–English language pairs were developed with the Marian toolkit (Junczys-Dowmunt et al., 2018). We developed models with both shallow and deep architectures, based on recurrent neural networks. We ensemble 4 independent runs and reranked with right-to-left models (output in reverse order). We saw

improvements with iterative back-translations and fine tuning on test sets from previous years, as well as use of the Paracrawl corpus (unfiltered).

A big challenge for system development are long training times (a month on a single GTX 1080ti GPU) which limited our ability to exploit the Paracrawl corpus. Because of this, we also started system development almost a year ago, using the training data from last year for the most part. All scores reported in this Section are on newstest2017 with case-sensitive BLEU.

2.1 Shallow System Development

We started with shallow systems similar to Edinburgh’s submission two years ago (Sennrich et al., 2016a). It uses byte pair encoding with a vocabulary of 50,000 (Sennrich et al., 2016c) and back-translation of the news2016 monolingual corpus (Sennrich et al., 2016b), about twice the size of the original training data.

For each training run, we compare different ways to obtain a single best model.

- Use the single model that performed best on the dev set (newstest2016).
- Use checkpoint ensembling to obtain the 4 or 8 best models, and decode the test set with an ensemble of these models.
- Merge the models obtained by checkpoint ensembling into a single model.

For German–English, we achieved slightly better results with an ensemble of independent models rather than a merged model (about +0.2 BLEU), while for English–German they perform similarly. Ensembling of either kind clearly outperforms the single best model.

We then built ensembles of the resulting systems for the 4 independent runs. This gives gains

¹Scores reported at <http://matrix.statmt.org/>